

Sara Rajabzadeh

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SUMMARY

Computer Vision Engineer with four years of research and engineering experience across medical imaging, image quality assessment, and 3D human pose forecasting. Skilled in Python, PyTorch, CNNs, diffusion models, uncertainty-based active learning, training on HPC clusters, rigorous benchmarking, and deployment with Docker and FastAPI. Co-authored research accepted at the NeurIPS 2022 Workshop and ICRA 2023, with experience spanning dataset curation, model development, evaluation, and production inference.

EXPERIENCE

Computer Vision Engineer Intern, Paris Brain Institute (ICM) Apr 2026 – Present

Supervisor: Dr. Gabriel Jimenez

- Developed an end-to-end deep learning pipeline for axon and myelin segmentation in transmission electron microscopy images, covering data preparation, manual annotation in Napari, nnU-Net v2 fine-tuning, inference, and post-processing on a 608-image dataset.
- Designed an iterative, multi-round active learning workflow combining nnU-Net v2 fine-tuning with entropy-based uncertainty sampling and successive manual annotation rounds in Napari.
- Built a multi-metric evaluation framework using Dice, HD95, and Panoptic Quality, and performed instance-level error analysis to identify disconnected myelin rings, over-segmentation, artifacts, and difficult pathological cases.
- Ran nnU-Net v2 preprocessing, fine-tuning, and inference on the institute's SLURM-based HPC cluster, including GPU job submission, environment configuration, and large-scale batch processing.
- Implemented morphological post-processing and watershed-based instance separation using scikit-image and SciPy, and reproduced and benchmarked AxonDeepSeg and nnU-Net segmentation baselines on microscopy datasets.

Computer Vision Engineer, Freelance Jun 2024 – Jan 2025

- Implemented and benchmarked multiple IQA and FIQA models, including TReS, OFIQ, GraFIQs, SDD-FIQA, and CR-FIQA, analyzing sensitivity to blur, noise, illumination, exposure, occlusion, head pose, and motion.
- Integrated RetinaFace and YOLOv8 face detection into FIQA pipelines and evaluated accuracy-latency trade-offs and the effect of face cropping on quality predictions.
- Fine-tuned TReS on natural and synthetic image-quality datasets and evaluated cross-dataset generalization on KADID-10k, KonIQ-10k, SPAQ, and CSIQ using SRCC and PLCC, identifying domain-shift and overfitting failure modes.
- Deployed a GPU-backed TReS inference service using FastAPI, Docker, CUDA, and Nginx, optimizing model initialization to load weights once and reuse the model across requests.
- Developed an image-quality metadata API for sharpness, dynamic range, illumination, and exposure, with unit and endpoint tests covering invalid inputs, multiple formats, image-size extremes, and runtime behavior.

Research Assistant, University College London Jul 2022 – Oct 2022

Supervisor: Dr. Evangelos Mazomenos

- Developed an EfficientNet-based deep learning pipeline for detecting necrotizing enterocolitis (NEC) from neonatal abdominal X-rays.
- Trained and evaluated the model on 62 patients using patient-level 2-fold cross-validation under limited-data conditions.
- Applied domain-specific augmentation and regularization to reduce overfitting on the small clinical dataset.
- Achieved a mean F1 score above 0.80 across cross-validation folds for NEC detection.
- Used Grad-CAM to interpret model predictions and qualitatively assess attention to clinically relevant image regions.

Research Assistant, EPFL Dec 2021 – Jul 2022
Supervisor: Prof. Alexandre Alahi

- Co-authored *A Generic Diffusion-Based Approach for 3D Human Pose Prediction in the Wild* published at ICRA 2023 and presented at the NeurIPS 2022 Workshop on Score-Based Methods.
- Co-developed a conditional denoising diffusion formulation for jointly repairing incomplete pose observations and forecasting future 3D human poses.
- Developed and trained models to handle partial occlusions, missing joints, missing frames, and noisy pose estimates using a temporally cascaded diffusion architecture.
- Evaluated the framework on Human3.6M, HumanEva-I, AMASS, and 3DPW using ADE and FDE, outperforming prior stochastic and deterministic baselines in clean and noisy-observation settings.

EDUCATION

Université Paris-Dauphine — PSL, Paris, France 2025 – 2026
Master 2 (M2) in Artificial Intelligence — Expected September 2026

University of Tehran, Tehran, Iran 2024 – 2025
Master 1 (M1) in Artificial Intelligence.

Sharif University of Technology, Tehran, Iran 2017 – 2022
B.Sc. in Computer Engineering.

PUBLICATION

- Saadatnejad S., Rasekh A., Mofayez M., Medghalchi Y., **Rajabzadeh S.**, Mordan T., and Alahi A., *A generic diffusion-based approach for 3D human pose prediction in the wild*, ICRA 2023; NeurIPS 2022 Workshop on Score-Based Methods.

HONORS & AWARDS

PR[AI]RIE–PSAI Excellence Scholarship, Université Paris-Dauphine — PSL 2025–2026

- Selected as one of 15 recipients for academic excellence and research potential in artificial intelligence.

SKILLS

Programming	Python, R, Java
Deep Learning & Computer Vision	PyTorch, TensorFlow/Keras, nnU-Net v2, OpenCV, TorchVision, scikit-image, Albumentations; CNNs, diffusion models, transfer learning, semantic and instance segmentation, image quality assessment, active learning
Medical & Bio Imaging	TEM, X-ray, DICOM, NIFTI; AxonDeepSeg, napari, ImageJ/Fiji
MLOps & Compute	Docker, FastAPI, REST APIs, CUDA, GPU inference, SLURM/HPC, Nginx, Git, Linux
Data, Experimentation & Evaluation	NumPy, Pandas, SciPy, Matplotlib, TensorBoard, Weights & Biases; Dice, IoU, HD95, Panoptic Quality, SRCC, PLCC

LANGUAGES

English Professional working proficiency; TOEFL iBT: 92
French Elementary proficiency (A1; currently studying)
Persian Native